

Binning Off

Each pixel has its own value of brightness. (value based on the image)

2	3	2	2	3	2
2	3	5	5	3	2
2	3	6	6	2	1
2	3	6	6	2	1
3	8	8	6	4	2
3	6	5	5	5	5

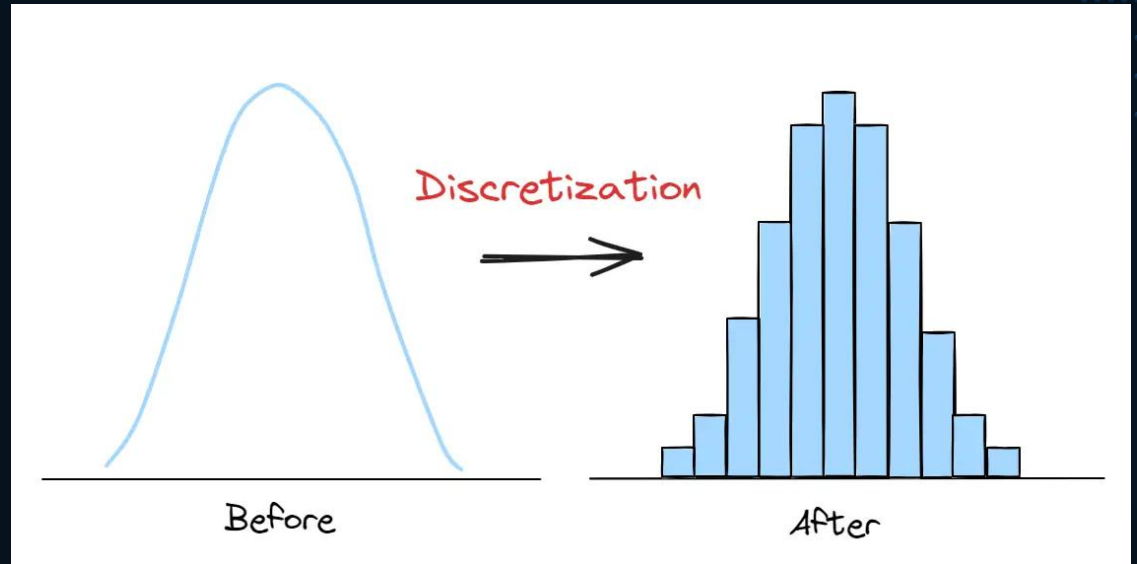


10	14	10
10	24	6
20	24	16

Binning On

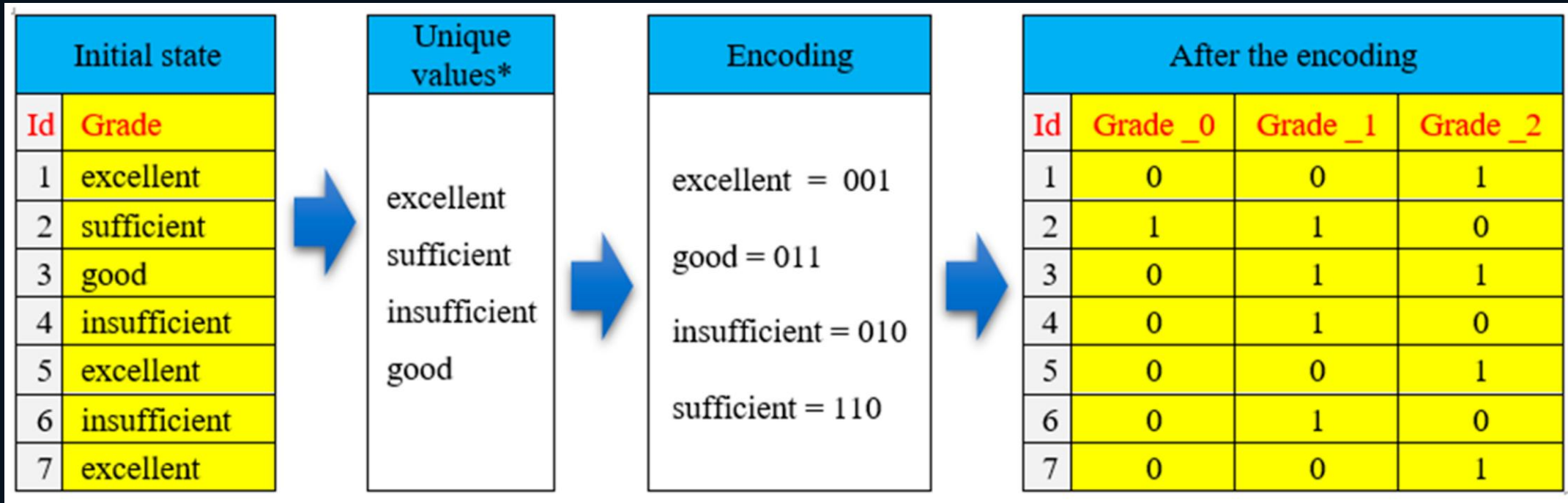
A 2*2 matrix of 4 pixels are grouped together and their individual values are combined. This combined value is substituted for original value.

SCALER
Topics



Binning & Discretization (From Scalars to Groups)

- **The Concept:** Reducing the "noise" of continuous numbers by grouping them into meaningful bins.
- **Why it works:** Sometimes the exact value (e.g., Age 22 vs 23) is less important than the category (e.g., Student vs Professional).
- **Types:**
 - **Equal Width:** Bins of equal size.
 - **Equal Frequency (Quantiles):** Bins with the same number of data points.



Encoding Categorical Data (Talking to the Machine)

- **The Problem:** Algorithms only understand math, not words like "Peshawar" or "Medical."
- **Label Encoding:** Assigning a unique integer to each category (0, 1, 2...). Best for "Ordinal" data where order matters (Small, Medium, Large).
- **One-Hot Encoding:** Creating a new binary column (0 or 1) for every unique category. Best for "Nominal" data where no order exists (City names).

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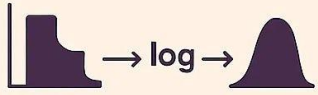


Feature Extraction (Synthesizing New Insights)

- **Definition:** Creating new variables from existing ones to expose hidden patterns.
- **Date/Time Extraction:** Breaking a "Timestamp" into 'Hour of Day', 'Day of Week', or 'Is Weekend'.
- **Mathematical Extraction:** Creating a 'Price per Square Foot' column from 'Total Price' and 'Area'.
- **Domain Specific:** In medical imaging, extracting the 'Aspect Ratio' of a detected cell.

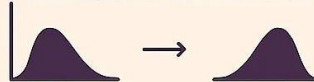
HANDLING SKEWED DATA IN DATA SCIENCE

LOG TRANSFORMATION



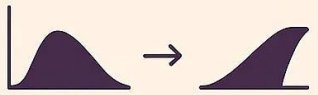
Apply the natural logarithm

SQUARE ROOT TRANSFORMATION



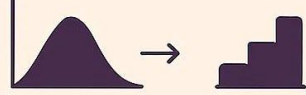
Apply the square root function

BOX-COX TRANSFORMATION



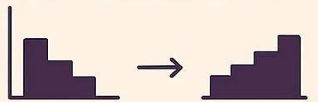
Find the optimal λ parameter

YEO-JOHNSON TRANSFORMATION



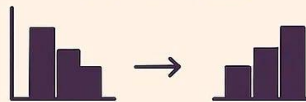
Like Box-Cox: works for zero negative values

BINNING (DISCRETIZATION)



Winsorizing, Clipping, Rank Transformation

OTHER TECHNIQUES



Rank Transformation

Handling Skewed Data (The Log Transform)

- **The "Bunching" Problem:** Real-world data (like wealth or rainfall) is often skewed, with most values bunched on the left and a few extreme values on the right.
- **Log Transformation:** Applying $\log(x)$ "pulls in" the extreme outliers and spreads out the bunched values.
- **The Result:** A distribution that looks "Normal" (Bell Curve), which is a requirement for many statistical models.